

Comparative fracture risk in vegetarians and nonvegetarians in EPIC-Oxford.

OBJECTIVE: To compare fracture rates in four diet groups (meat eaters, fish eaters, vegetarians and vegans) in the Oxford cohort of the European Prospective Investigation into Cancer and Nutrition (EPIC-Oxford). **DESIGN:** Prospective cohort study of self-reported fracture risk at follow-up. **SETTING:** The United Kingdom. **SUBJECTS:** A total of 7947 men and 26,749 women aged 20-89 years, including 19,249 meat eaters, 4901 fish eaters, 9420 vegetarians and 1126 vegans, recruited by postal methods and through general practice surgeries. **METHODS:** Cox regression. **RESULTS:** Over an average of 5.2 years of follow-up, 343 men and 1555 women reported one or more fractures. Compared with meat eaters, fracture incidence rate ratios in men and women combined adjusted for sex, age and non-dietary factors were 1.01 (95% CI 0.88-1.17) for fish eaters, 1.00 (0.89-1.13) for vegetarians and 1.30 (1.02-1.66) for vegans. After further adjustment for dietary energy and calcium intake the incidence rate ratio among vegans compared with meat eaters was 1.15 (0.89-1.49). Among subjects consuming at least 525 mg/day calcium the corresponding incidence rate ratios were 1.05 (0.90-1.21) for fish eaters, 1.02 (0.90-1.15) for vegetarians and 1.00 (0.69-1.44) for vegans. **CONCLUSIONS:** In this population, fracture risk was similar for meat eaters, fish eaters and vegetarians. The higher fracture risk in the vegans appeared to be a consequence of their considerably lower mean calcium intake. An adequate calcium intake is essential for bone health, irrespective of dietary preferences. **SPONSORSHIP:** The EPIC-Oxford study is supported by The Medical Research Council and Cancer Research UK.